LS-CAT Management Board Meeting

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May 16, 2014



LS-CAT Management Board Meeting Agenda

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Start Time: 1 PM CDT (2 PM EDT) May 16, 2014
     Audio: 1-888-850-4523
  Passcode: 190580
   Video 1: http://conf1.ls-cat.org
   Video 2: http://conf2.ls-cat.org
Physical Location: 9700 S. Cass Ave Bldg 436A, Argonne, IL
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- Approval of last meeting's minutes
- Status
  - LS-CAT Contract Extension
  - ANL top down changes
  - Repolished 21-ID-D mirrors installed
  - New data collection software and remote client rollout
- Data Collection/Analysis Workshops @ LS-CAT
- Wayne State
- Budget Guidance
- Budget (Flat budget proposed, **2 votes**)
- Capital fund: Detector time? (**1 vote**)
- APS/SAC Review of LS-CAT November 4, 2014: Board meeting / symposium November 3?
- New Business

LS-CAT Extension

Northwestern has expressed support for a 5 year extension of the current contract with the caveat that the price per share not increase beyond inflation.

Northwestern has not yet proposed the specifics of the contract amendment that would implement the extension.

If accepted by the member institutions the new expiration date would be February 28, 2022.

As discussed previously, the out years would (hopefully) overlap with a capital improvement project and transition to a post multi bend achromat era with the ramp down of the LS-CAT budget and ramp up of an LS-CAT-2 budget.

Although the APS party line is that there will be a ~12 month rebuilding period in 2019 there is no proposal on the table that specifies either that length or that year.

ANL/APS/Upgrade Changes Since Last Meeting

Peter Littlewood succeeds
Eric Isaacs as ANL director





Stephen Streiffer named interim APS Director replacing
Brian Stephenson

Stuart Henderson succeeds George Srajer
as APS Upgrade Director

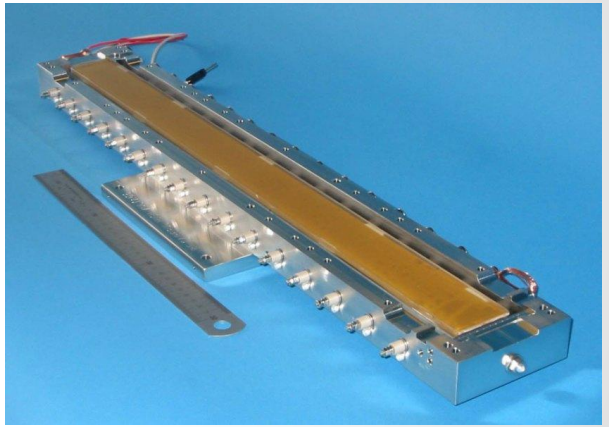


LS-CAT Status Update

Repolished and recoated mirror installed on 21-ID-D, will see beam this coming run.

Mirror mechanics were not replaced as the advantages of doing so were not clear.

Vibration isolation now being considered: much less expensive than new mechanics and much more likely to work. Downside? Moving a 3 ton granite block that has adhered to the floor will be a treat.



New Data Collection Software

Motivation for drastic software overhaul:

Current MD-2 software (and computer hardware) is obsolete.

Current remote access client uses Adobe Flash:

Although once ubiquitous, it is now rapidly becoming obsolete.

The underlying communication protocol our remote access client uses is at the limit of what we can do with it.

We need to update our experimental capabilities.

New Data Collection Software

Remote access client rewritten using HTML5 and javascript to allow remote use without having to download or install special programs.

Status information now handled internally using a non-SQL database to avoid blocking queries.

Control information continues to be handled by our SQL database to provide security and consistency.

Major New Features

Helical and Raster scans supported without the (same) problems and bugs that plague the current “beta” code.

Only one person can control the station at a time.

- Users must be listed as “On-Site” or “Remote” on the ESAF.
- Staff can take control away from the remote user at any time.

Faster centering than the current remote client.

Local users will use the same interface as the remote users.

New Software Rollout Plan

2014-2: Allow some experimenters to use the new software on at least one station. Use feedback from users and staff to improve the experience.

2014-3: Run mainly with the new software maintaining the old systems as a backup.

2015-1: Virtualize the XP computers used to run the old computers and decommission the hardware. This is the point of no return as the (obsolete) video framebuffers used cannot be virtualized. However, some XP only software used to maintain the MD2s will still run fine.

LS-CAT
Life Sciences Collaborative Access Team

kebrister

Schedule | 21-ID-D | 21-ID-E | 21-ID-F | 21-ID-G

112039 Brister | Log Out | You are controlling this station | Cede Control

Sample | Align | Collect | MD2

Align: Move along a line segment

Beam Size: 20 µm dia circle

6 Click Crystal Alignment

Begin: Get Movie

Movie Progress

Select Starting Point: Accept

Select Ending Point: Accept

Simulation Range:

Zoom: 1 2 3 4 5 6 7 8 9 10 | Retake Movie with Same Zoom | Retry

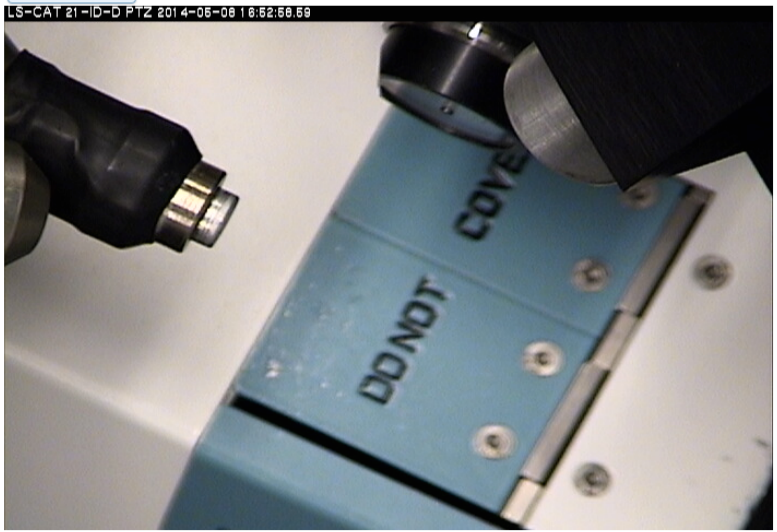
Instructions

This is similar to normal 3-click alignment EXCEPT that you are choosing two points on your cyrstal instead of one. As the data collection progresses the crystal will be translated continuously from the first point to the second. This is also called "vector" or "helical" data collection.

For both the starting point and the ending point click on your cyrstal at three different angles. After the 3rd click a yellow spot showing where the beam will hit your crystal will appear. If you like the result press the "accept" button. To try again or to refine your selection at higher magnification hit "retry" or select a new zoom level, respectively.

The angle you choose for the first click for the starting point will be 0° on the data

D Mount Area



Challenges

We have been using the current systems for many years now. We have many procedures that rely on our knowledge of the strengths and weaknesses of what we've been using. We need to roll out the new software in a way that allows us to relearn how to run the beamlines without adversely effecting user beamtime.

2014-1 LS-CAT Data Collection and Structure Solution Workshop

- On April 14 & 15 LS-CAT hosted a user workshop
- 9 Participants
- 1st day covered
  - Data collection theory
  - Strategy theory
  - Processing programs
  - Remote Access
  - Data collection at the beamlines w/test xtals & student's projects
- 2nd day covered
  - General overview or structure determination pipelines
  - Data processing cutoffs & recent literature
  - Hands-on computer tutorials working with data at the beamlines

Future Workshop Plans

- Have more workshops on this scale, keep it small and more interactive
- Next workshops will be Aug 11 & 12 and Oct 13 & 14
- Encourage members to apply to APS CCP4 workshop

Wayne State

WSU terminated its relationship with LS-CAT effective June 10, 2014. However, discussions have continued and there is a possibility they would want to rejoin at a reduced level after June 16.

How would we like to handle a full share member wishing to go to a half share?

Should any member be allowed to reduce their share by giving one year's notice?

Budget Guidance

Several people have expressed concern that if we lose a member that this will mean an automatic increase in the dues. Unless the LS-CAT management board directs me to do otherwise, I will be proposing budgets that attempt to maintain a nearly constant per share rate, possibly adjusted for inflation.

Consequences:

1) Increasing the number of active LS-CAT shares will increase the level of service or increase the rate at which detectors and other beamline equipment can be upgraded.

2) Decreasing the number of active LS-CAT shares will require a corresponding reduction in service.

LS-CAT FY 15 Proposed Budget

Description	FY2014 Total (Est)	FY2014 Budget	FY15 FLAT
Academic Personnel	452,436	452,436	466,009
Non-Academic Personnel	63,024	110,929	64,915
Fringe Benefits	225,799	251,146	236,872
Salaries Restricted	315,020	314,768	324,211
Supplies	32,051	76,000	36,000
Supplies, Restricted	2,411	2,000	2,500
Services	21,522	13,000	27,000
Services, Restricted	17,733	17,000	18,000
External Space Rental	113,233	113,233	116,630
Travel, Domestic	11,275	5,000	10,000
Utilities & Telephone	8,112	8,000	13,200
Capital Equipment	256,155	256,155	13,100
Indirect Cost - Sponsored	292,514	388,833	371,064
FY14 WSU Shortfall		-51,500	
Totals:	1,811,284	1,957,000	1,699,500
Per Share	278,659	309,000	309,000

So far in FY14 we've been able to keep costs under control and live within our means. Spending for equipment repairs has been much less than anticipated and with the reduction in staff it looks like we'll come in under budget even with the WSU shortfall.

At this point we can anticipate that by the end of FY14 (August 31, 2014) our capital fund will be about $2.0M.

The proposed flat budget for FY15 has a ridiculously small amount for the capital equipment line. Unless some or all of the WSU share is picked up we will definitely be dipping into the capital fund to maintain the facilities.

Budget FY 15

Salaries			
Base	885,135		100% KB, JSB, MB, JVO, DS, SA, ZW
Fringe	236,872		95% NB
Total		1,092,007	
Items w/o F&A			
Capital Eq.	13,100		
Rent	116,630		
Telcom	13,200		
Total		142,930	
Items w/F&A			
Lab Supplies	36,000		
Office Supplies	2,500		
Lab Services	27,000		
Insurance	18,000		
Travel	10,000		
Total		93,500	
F&A		371,064	
Total		1,699,500	
Per Share		309,000	⇐ NU, UM, MSU, VARI
Per 0.5 Share		154,500	⇐ UI, VU, UW

Vote 1:
Approval of amounts shown. In the event the WSU share is reassigned, either fully or partially, the amount billed will increase the capital equipment budget line.

Vote 2:
The capital equipment line will be used for maintenance whenever appropriate. In the event required maintenance exceeds the budget, the saved capital equipment fund will be used to make up the difference. The board will be kept informed about unusually large expenses.

Uses for Capital Fund: What gives the most value to the membership?

How can we use the capital fund to make LS-CAT as attractive as possible to potential new members?

Upgrade a detector. ($1.6M - $2.0M)

Support unipuck and offer faster sample exchanges. (~$100K?)

Bring 21-ID-E online. (~$500K?)

Suspend dues for a year.

Hold as a contingency fund.

New Detector status

The NIH proposal for a Pilatus3 6M was not successful. Purchasing this particular detector on our own is not a great idea as it would completely deplete our capital reserves. However, the Rayonix HS series detectors are now being delivered and the first reports are quite positive. At under $1.6M (compared to ~$2M for the Pilatus3 6M) these detectors are conceivably in our price range.

MX-HS Technical Specifications

Sensors	Proprietary Rayonix frame-transfer CCD
Dead Time	1 millisecond
Full Well Capacity, at 2 x 2 Binning	400ke⁻/pixel
Noise	High Speed mode: 8 e⁻/pixel Low Noise mode: 4 e⁻/pixel
Dynamic Range	High Speed mode: 16 bit Low Noise mode: 18 bit
Dark Current	0.003 e⁻/pixel/second or 0.0004 photons/pixel/second (12keV)
Phosphor	40µm standard, 20–80µm by special order
PSF, FWHM	100µm with 40µm thick phosphor, 65µm with 25µm phosphor
Sensor Operating Temperature	–80° C
Cooling	Closed-cycle refrigeration

User-configurable Imaging Parameters

	rayonix mx170 HS	rayonix mx225 HS rayonix mx300 HS	rayonix mx340 HS rayonix mx425 HS
On-chip Binning	**Pixel Size (µm)**	**Pixel Size (µm)**	**Pixel Size (µm)**
1 x 1	44	39	44
2 x 2 *(standard)*	89	78	89
3 x 3	133	117	133
4 x 4	177	156	177
5 x 5	221	195	221
6 x 6	266	234	266
8 x 8	354	312	354
10 x 10	440	390	440
Other Specifications			
Fiber-optic Taper Demagnification Ratio	2.92:1	2.6:1	2.92:1
Electro-optical Gain	7e⁻/12keV photon	9e⁻/12keV photon	7e⁻/12keV photon

rayonix mx170 HS	rayonix mx225 HS	rayonix mx300 HS	rayonix mx340 HS	rayonix mx425 HS
4 Modules 15 Megapixels	9 Modules 33 Megapixels	16 Modules 59 Megapixels	16 Modules 59 Megapixels	25 Modules 92 Megapixels
170 mm x 170 mm, 28,900 mm²	225 mm x 225 mm, 50,625 mm²	300 mm x 300 mm, 90,000 mm²	340 mm x 340 mm, 115,600 mm²	425 mm x 425 mm, 180,625 mm²
64 channels	144 channels	256 channels	256 channels	400 channels
34 cm x 30 cm x 40 cm	46 cm x 37 cm x 40 cm	58 cm x 47 cm x 43 cm	58 cm x 47 cm x 43 cm	68 cm x 57 cm x 43 cm
30 kg	57 kg	120 kg	130 kg	200 kg
175 cm x 64 cm x 64 cm	175 cm x 64 cm x 64 cm	175 cm x 64 cm x 64 cm (x2)	175 cm x 64 cm x 64 cm (x2)	175 cm x 64 cm x 64 cm (x3)
215 kg	215 kg	215 kg (x2)	215 kg (x2)	215 kg (x3)

New Detector Vote

Rayonix has started to deliver new HS series detectors. While these first detector were quite late this is in keeping with previous detectors from this company: the initial detectors were delayed while production ramped up and the subsequent detectors were delivered on time.

Vote:

1) Get new quotes from ADSC, Dectris, and Rayonix. If Rayonix MX 300 HS has strong competition at the $1.6M price point then advise board, otherwise order Rayonix MX 300 HS.

2) Let's form a committee to review the detector situation.

3) No, let's not buy a new detector at this time.

APS SAC Review of LS-CAT

The APS SAC will review LS-CAT on Tuesday November 4, 2014.

Possible schedule: have a symposium (perhaps in Evanston) and board meeting on Monday November 3 followed by the review itself.